

Networking Services: NAT,



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Network Address Translation - NAT



Private Network

- *Private IP* network is an IP network that is not directly connected to the Internet
- Not e
- **All** appear to have the same IP to the outside world
- Generally, private networks use addresses from the following experimental address ranges (*non-routable addresses*):
 - 10.0.0.0 – 10.255.255.255
 - 172.16.0.0 – 172.31.255.255
 - 192.168.0.0 – 192.168.255.255

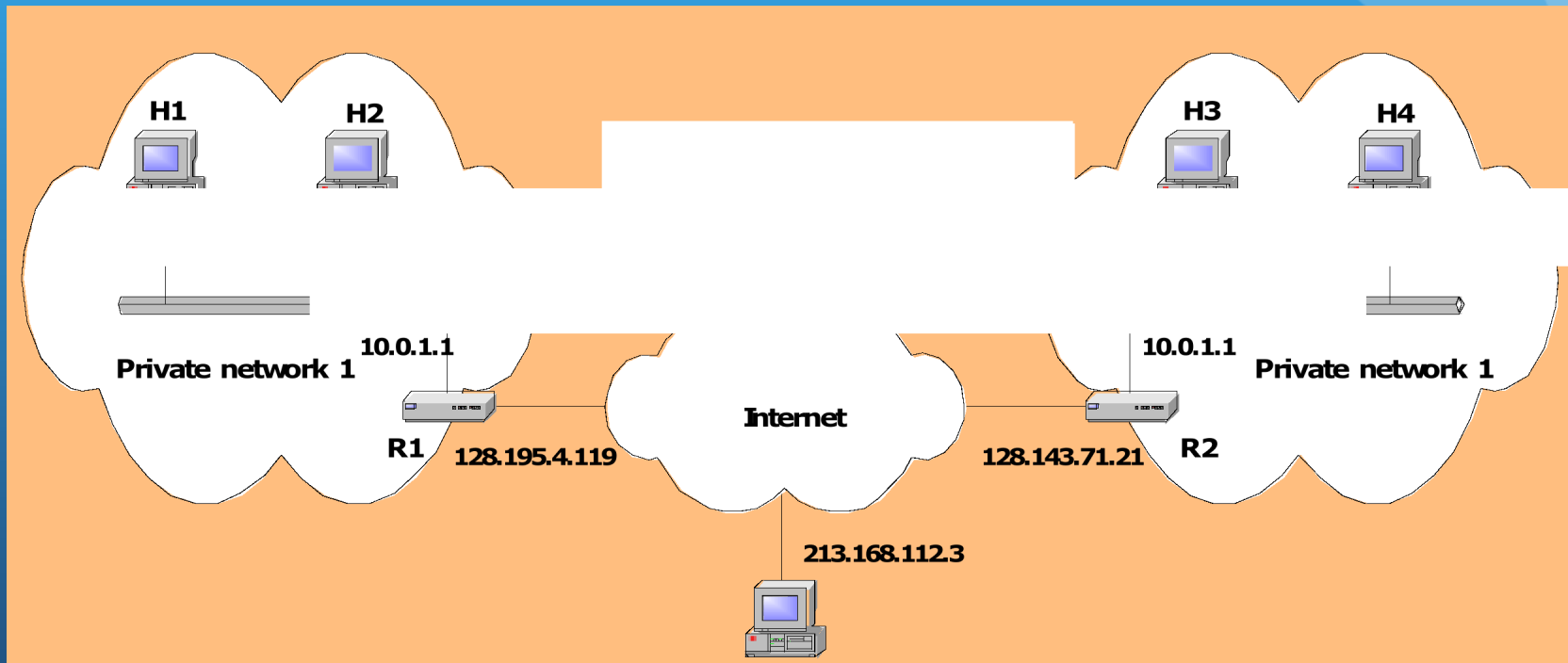
Implication of NATs

- NAT breaks one of the fundamental assumption of Internet: that all machines are peers and are routable by IP number.

As such:

- [REDACTED]
- [REDACTED]
- NAT uses
 - The NAT needs to discover or be told that port 80 (web service) packets need to be routed to a specific machine
 - Most home gateways have functionality for this specifically for running (web, game) servers!
- This is a problem for any peer to peer system. Your likely experience with it is using Skype (discuss later)

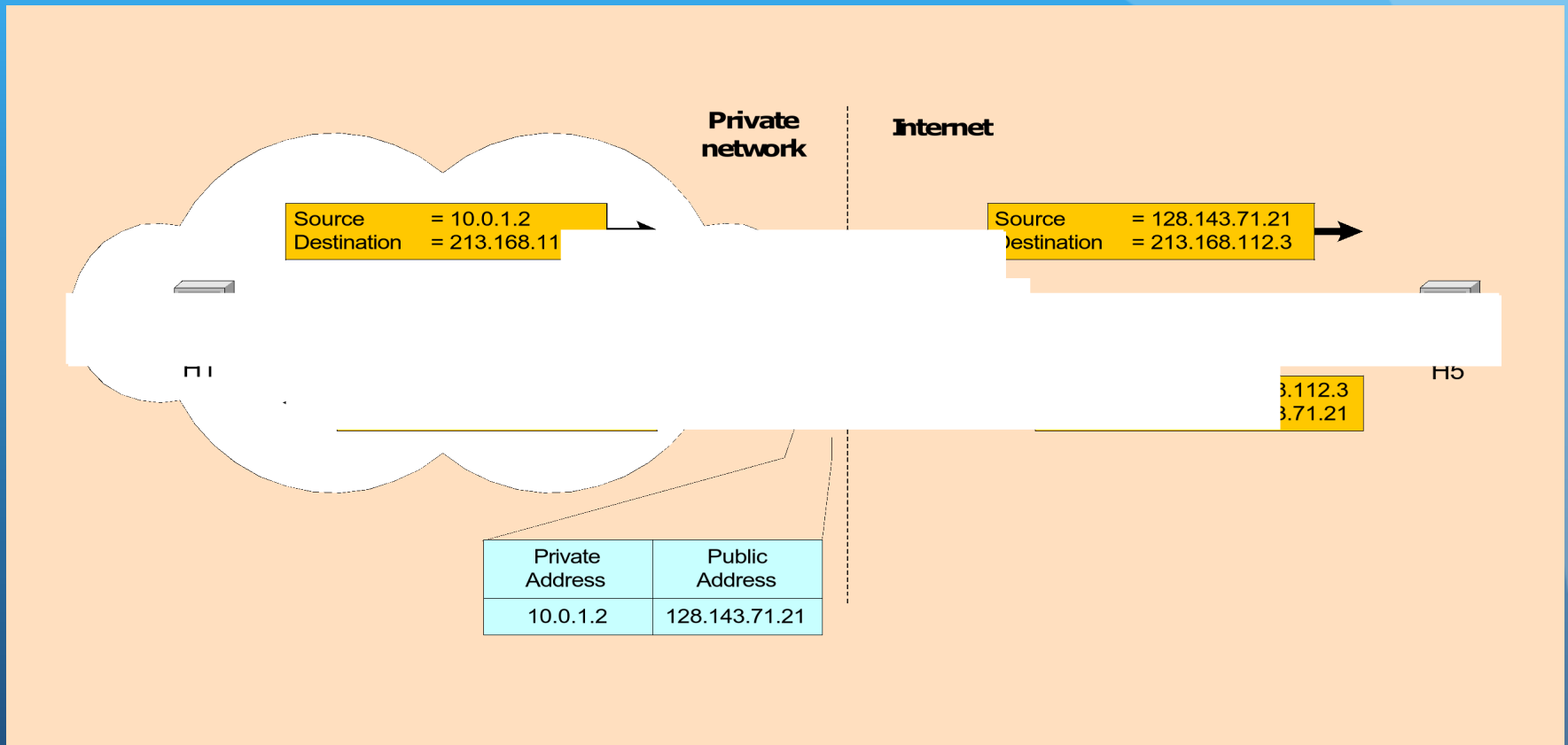
Private Addresses



Network Address Translation (NAT)

- NAT is a router function where IP addresses (and possibly port numbers) of IP datagrams are replaced at the boundary of a
- NAT i _____ networks to communicate with hosts on the Internet
- NAT is run on routers that connect private networks to the public Internet, to replace the IP address-port pair of an IP packet with another IP address-port pair.

Basic operation of NAT

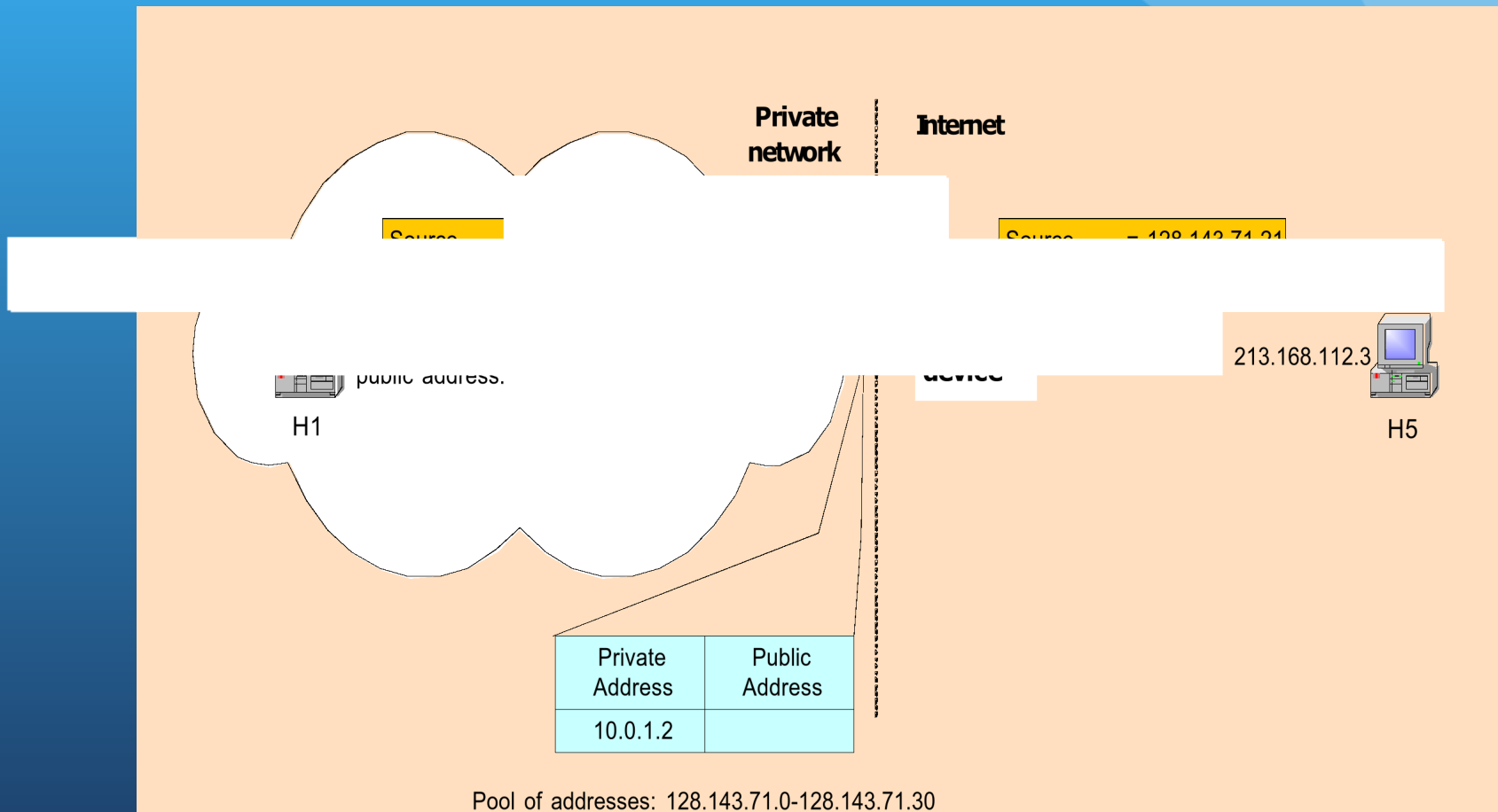


- NAT device has address translation table
- One to one address translation

Pooling of IP addresses

- **Scenario:** Corporate network has many hosts but only a small number of public IP addresses
- NAT device in corporate network and the public Internet, manages a pool of public IP addresses
- When a host from the corporate network sends an IP datagram to a host in the public Internet, the NAT device picks a public IP address from the address pool, and binds this address to the private address of the host

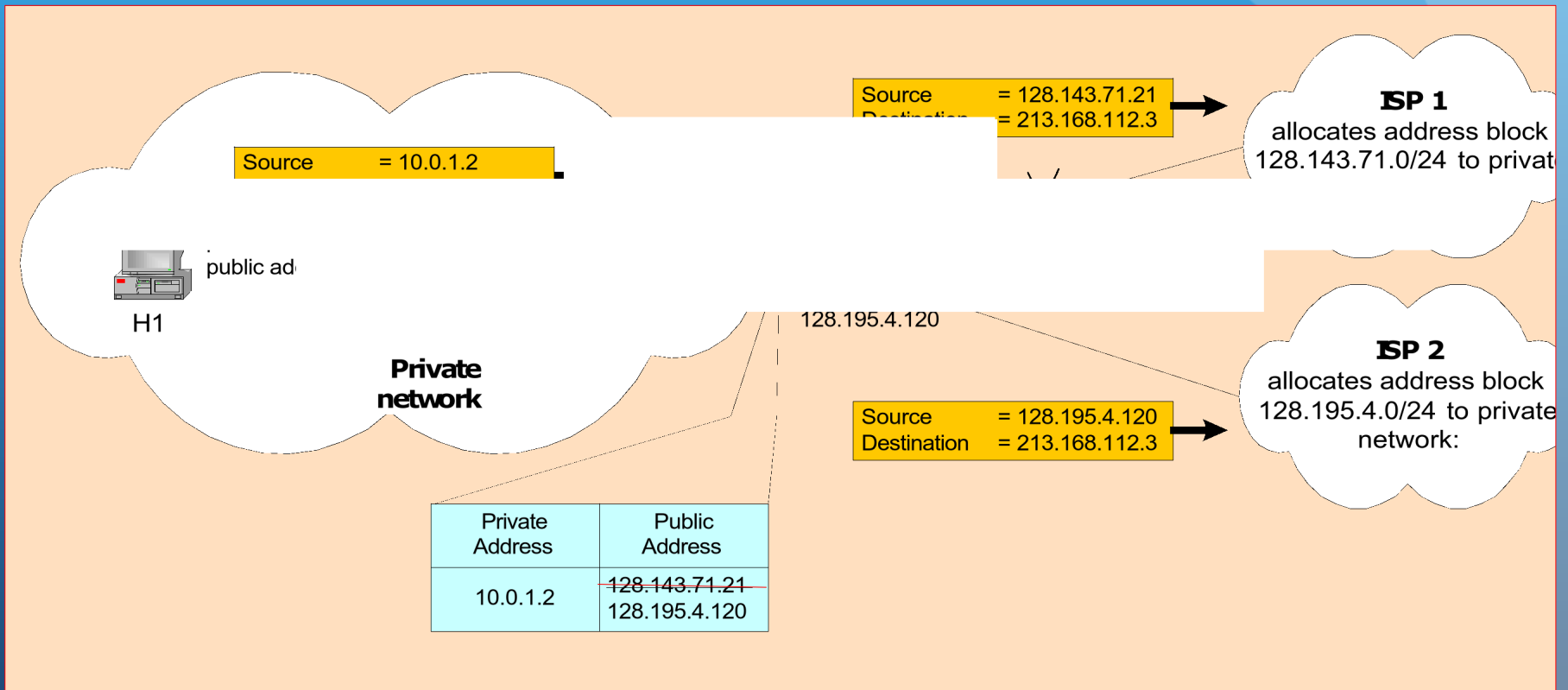
Pooling of IP addresses



Supporting migration between network service providers

- **Scenario:** In CIDR, the IP addresses in a corporate network are obtained from the service provider. Changing the service provider requires changing all network.
 - NAT bind the private address of a host to the public address.
 - Migration to a new network service provider merely requires an update of the NAT device. The migration is not noticeable to the hosts on the network.
- Note:**
- The difference to the use of NAT with IP address pooling is that the mapping of public and private IP addresses is static.

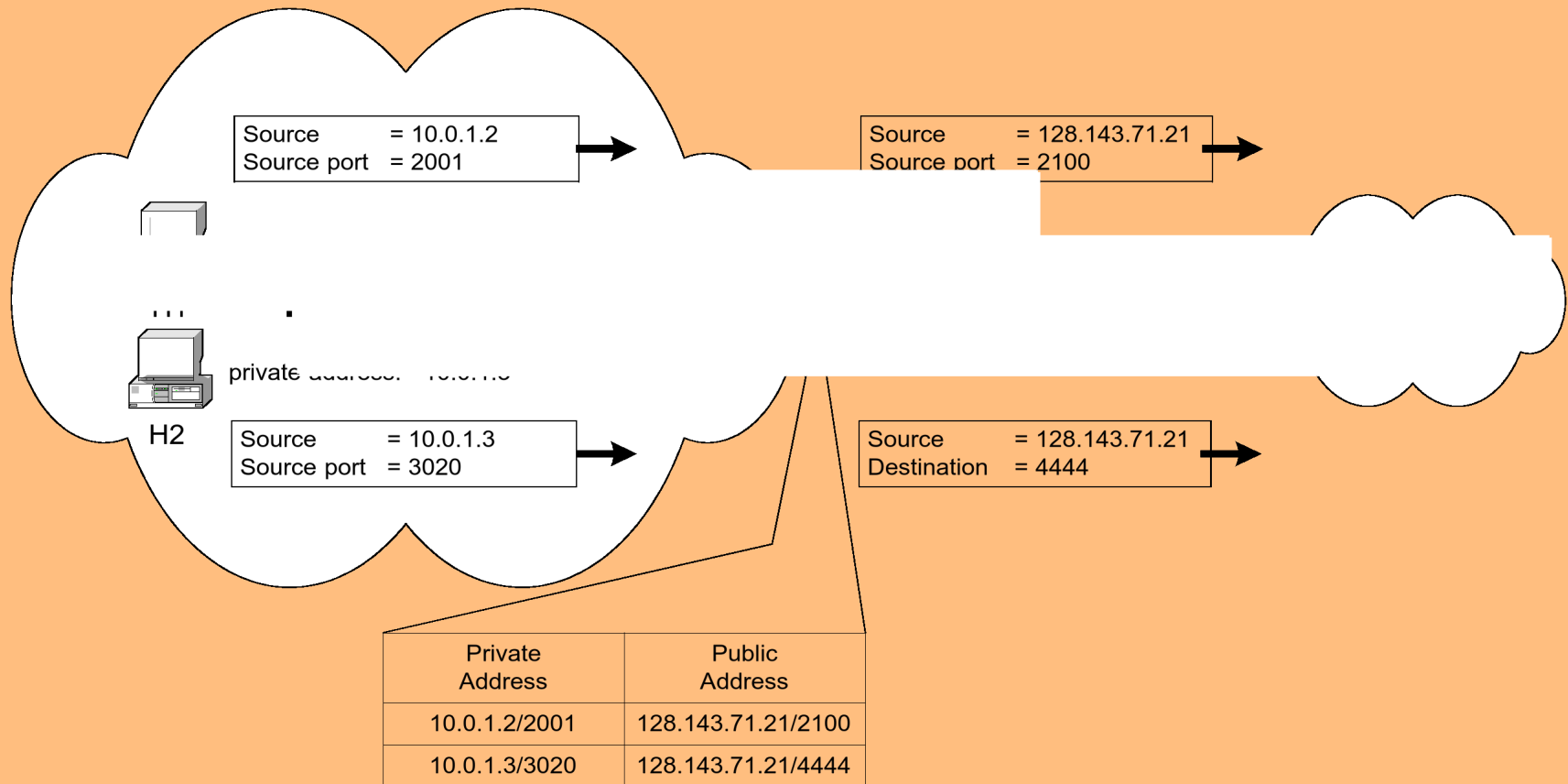
Supporting migration between network service providers



IP masquerading

- **Also called: Network address and port translation (NAPT), port address translation (PAT).**
- **NAT solution:**
 - Assign private addresses to the hosts of the corporate network
 - NAT device modifies the port numbers for outgoing traffic

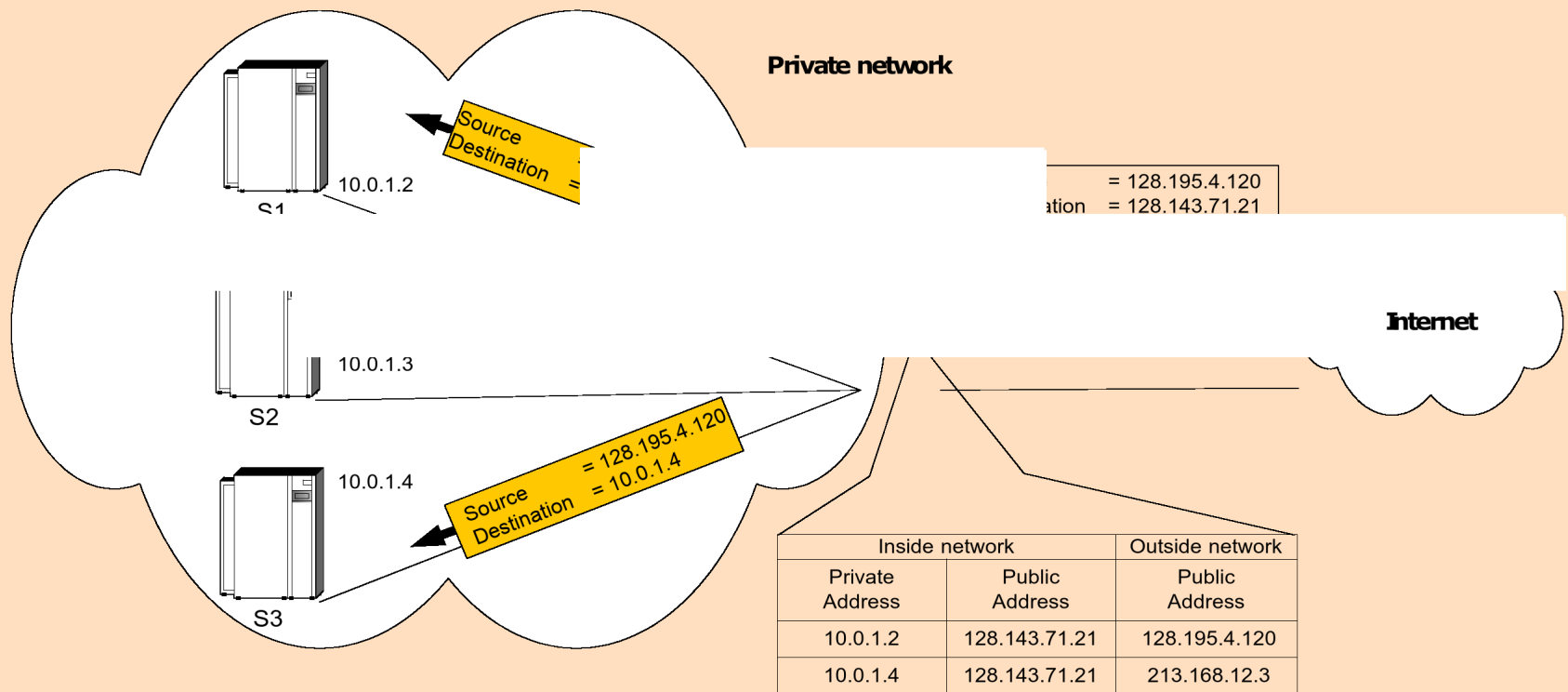
IP masquerading



Load balancing of servers

- **Scenario:** Balance the load on a set of identical servers, which are accessible from a single IP address
- Her
- NAT device acts as a proxy for requests to the server from the public network
- The NAT device changes the destination IP address of arriving packets to one of the private addresses for a server
- A sensible strategy for balancing the load of the servers is to assign the addresses of the servers in a round-robin fashion.

Load balancing of servers



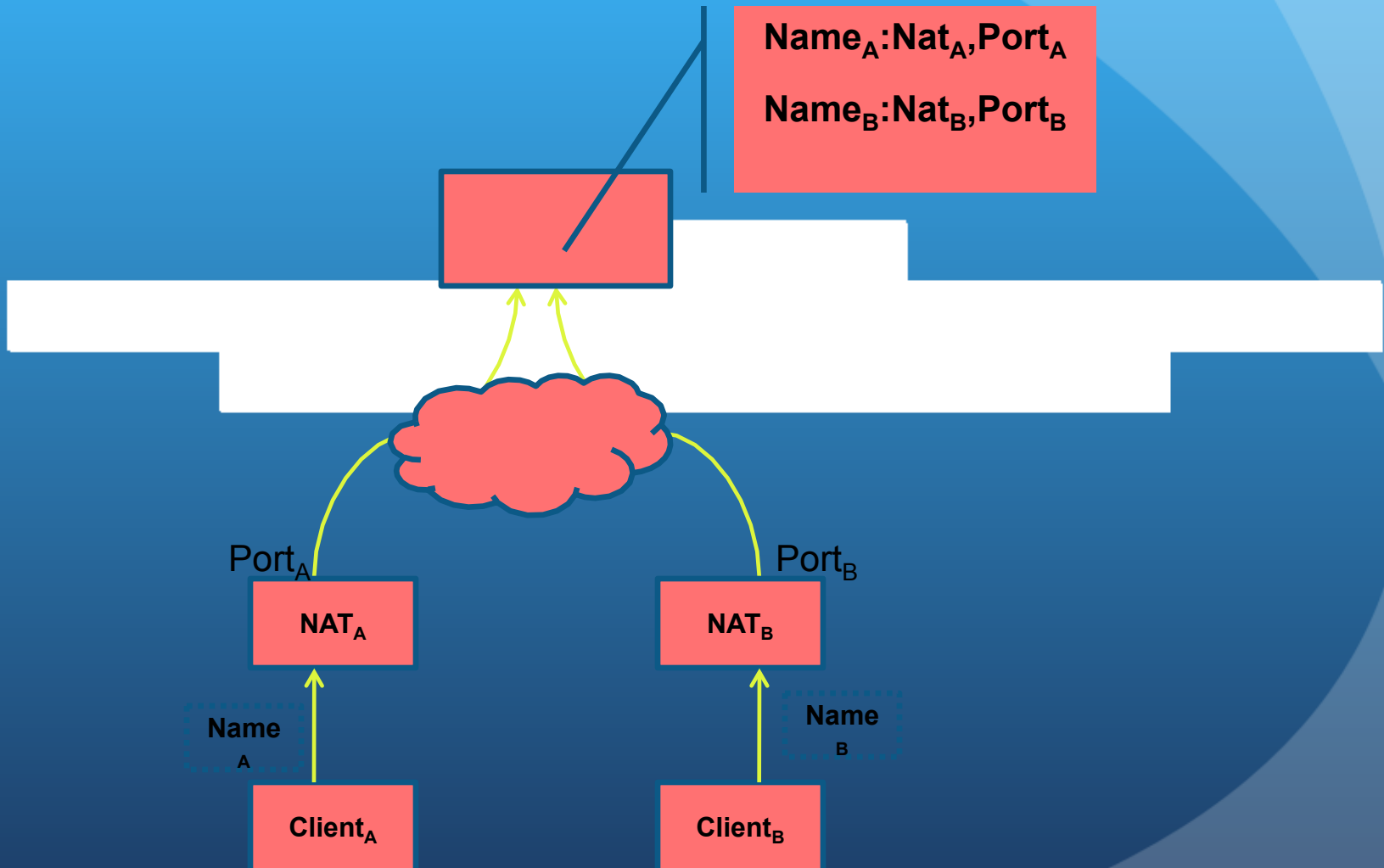
Concerns about NAT

- **Performance:**

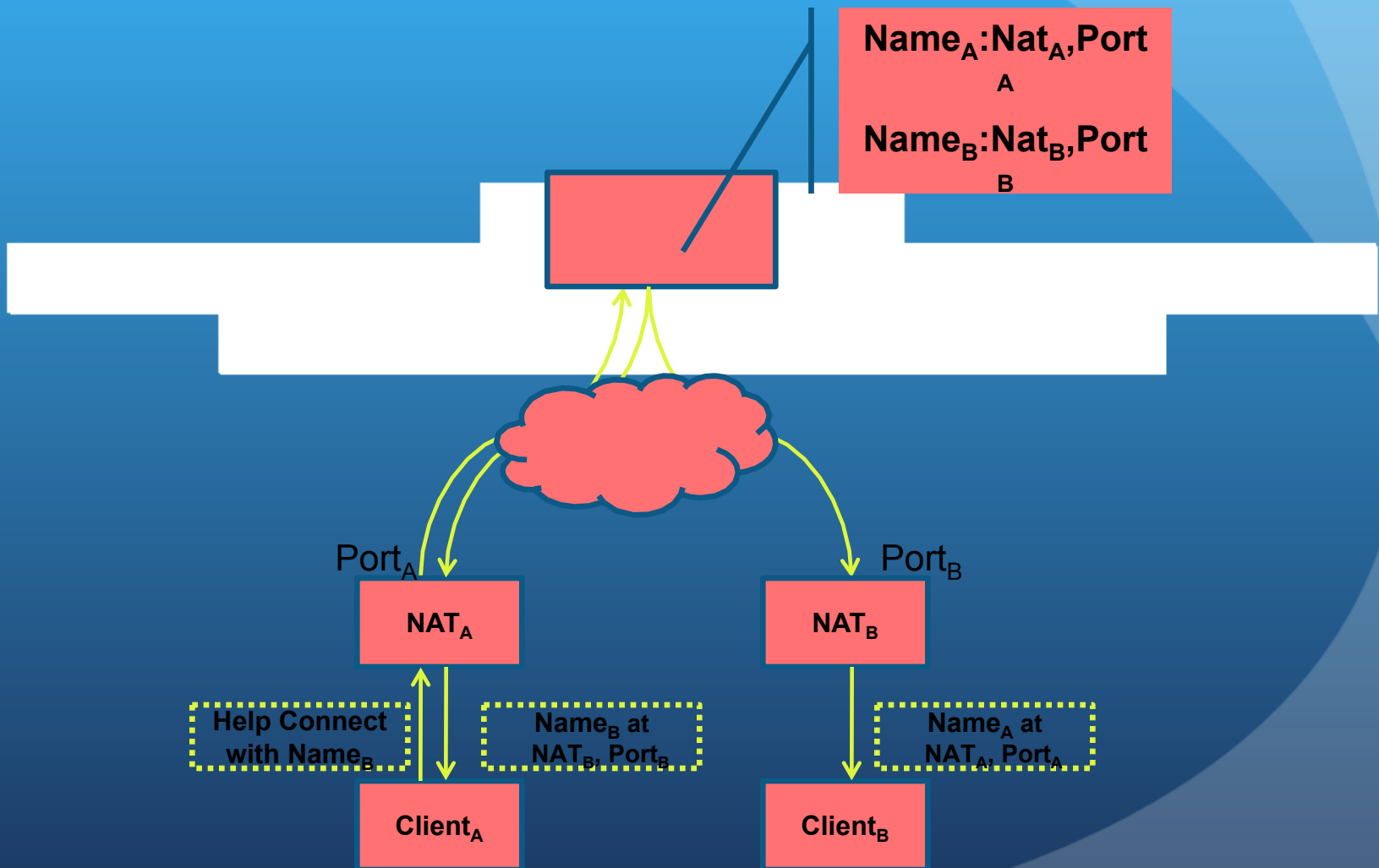
- Modifying the IP header by changing the IP address requires that NAT boxes recalculate the IP header checksum
- Modifying port number (and sequence number) requires that NAT boxes recalculate TCP and UDP checksums (and IP header)

- NAT is not symmetric. A host in a private network cannot initiate communication to a host in the public Internet.
- A host in the public Internet often cannot initiate communication to a host in a private network unless permanently mapped in table.
- The problem is worse, when two hosts that are in a private network need to communicate with each other. Peer to Peer
 - Solution to that is using NAT traversal.
 - Skype uses that but a server is used to relay the messages between clients.

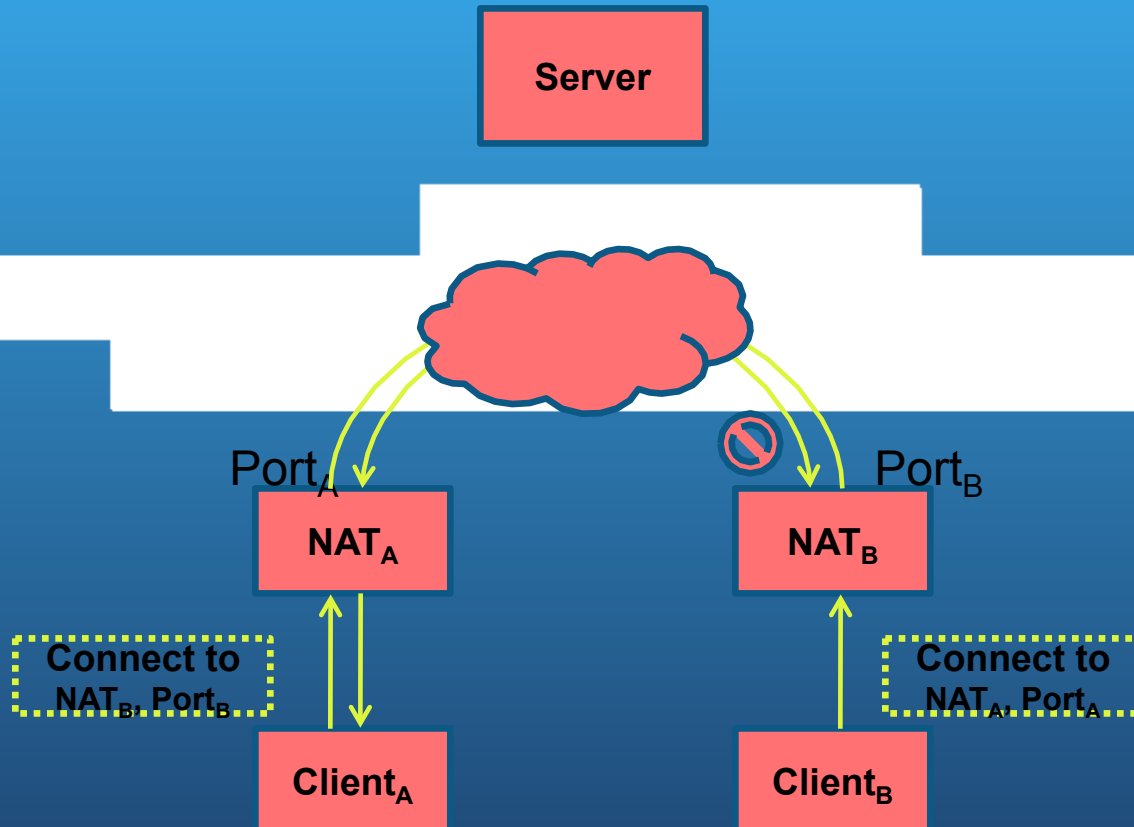
UDP Hole Punching



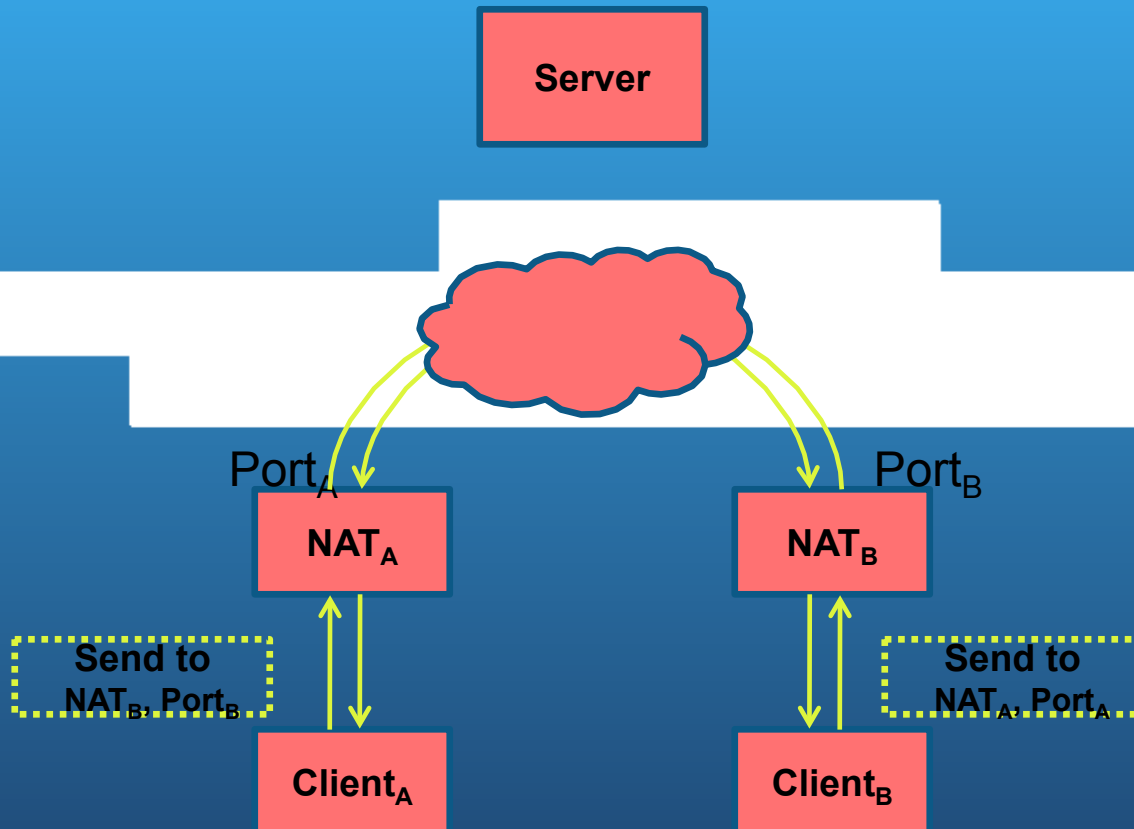
UDP Hole Punching



UDP Hole Punching



UDP Hole Punching



Concerns about NAT

- **IP address in application data:**

- Applications that embed IP addresses in the payload of the application data are not supported by NAT. This is a problem for applications that use a private-IP address to connect to a server.
 - So, applications that use a private-IP address to connect to a server are not supported by NAT. This is a problem for applications that use a private-IP address to connect to a server.
- application layer protocols and, if an IP address is detected in the application-layer header or the application payload, translate the address according to the address translation table.

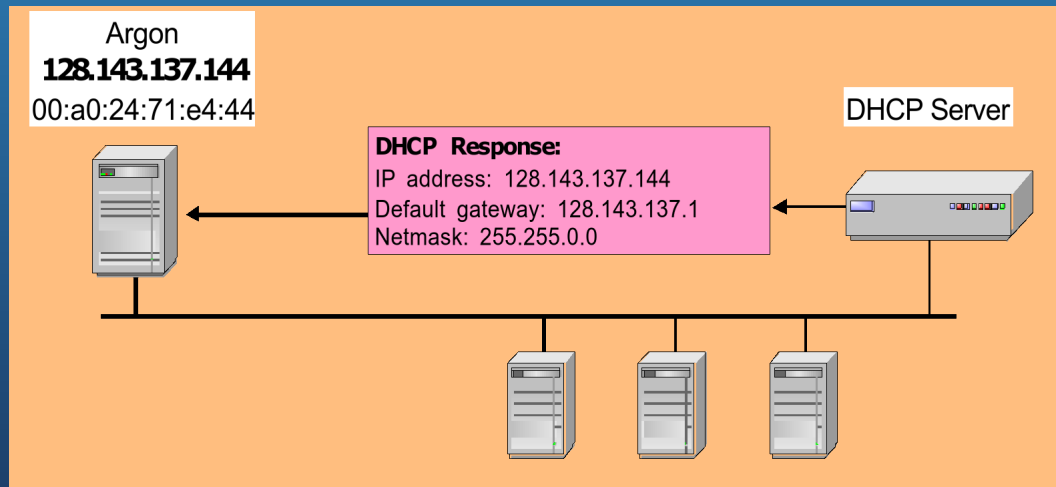
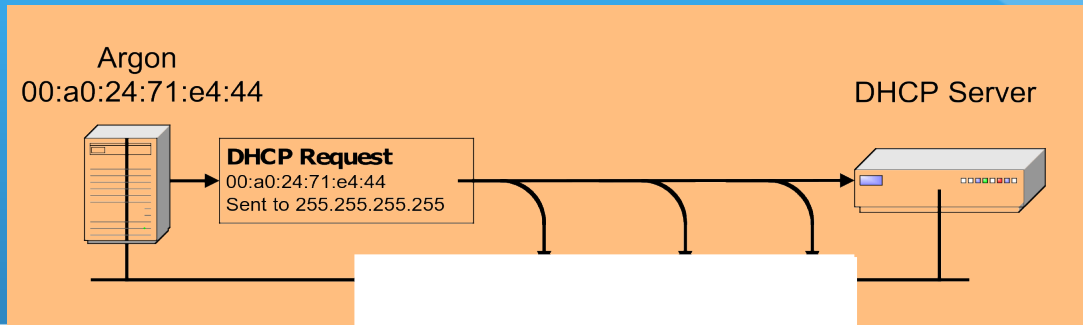
Dynamic Host Control Protocol - DHCP



Dynamic Assignment of IP addresses

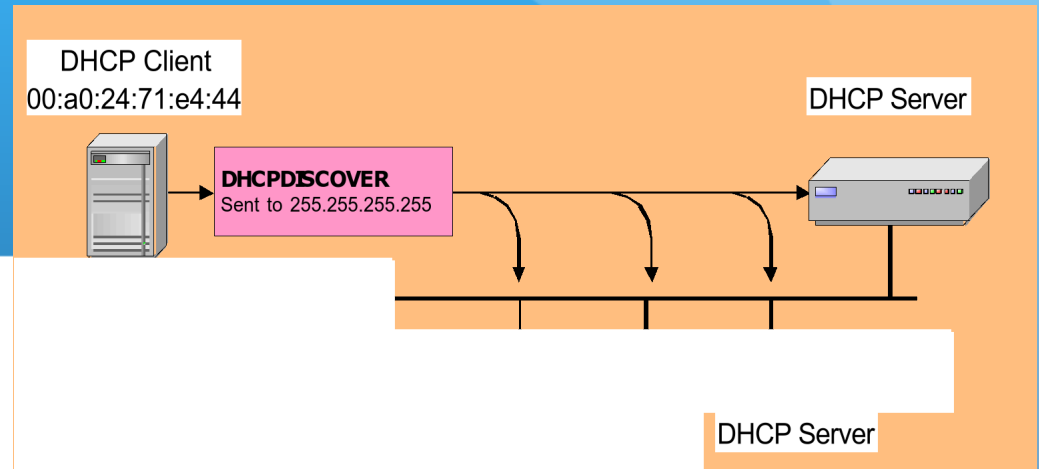
- Dynamic assignment of IP addresses is desirable for several reasons:
 - IP addresses are
 - Sup

DHCP Interaction (simplified)

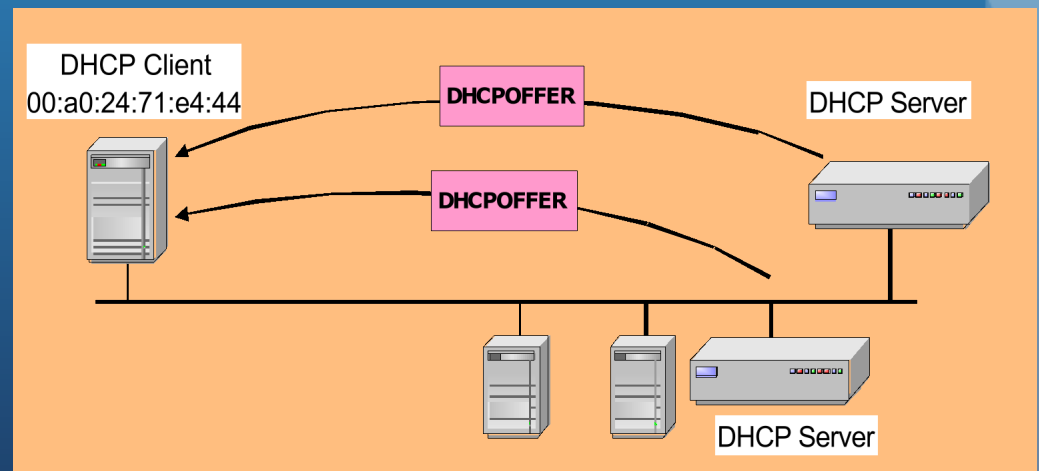


DHCP Operation – First search for DHCP servers

- DCHP DISCOVER



- DCHP OFFER



Client-Server Interactions

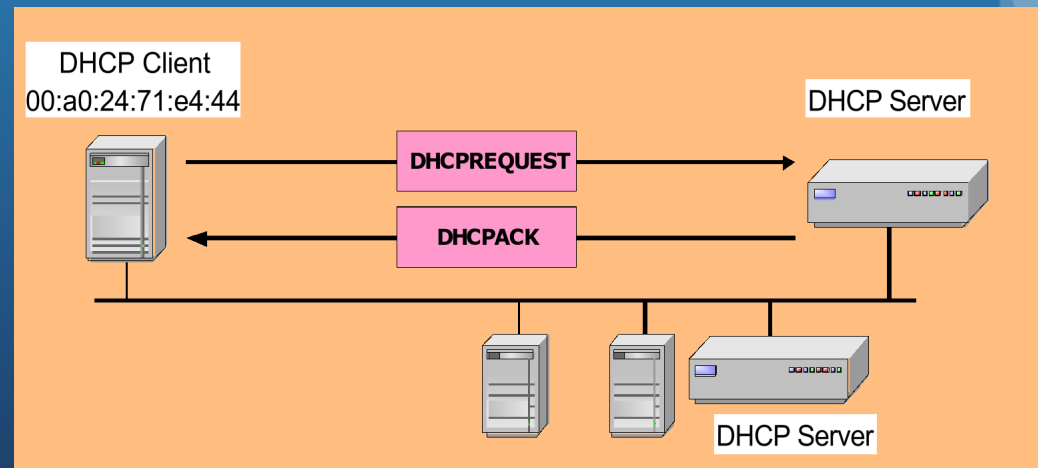
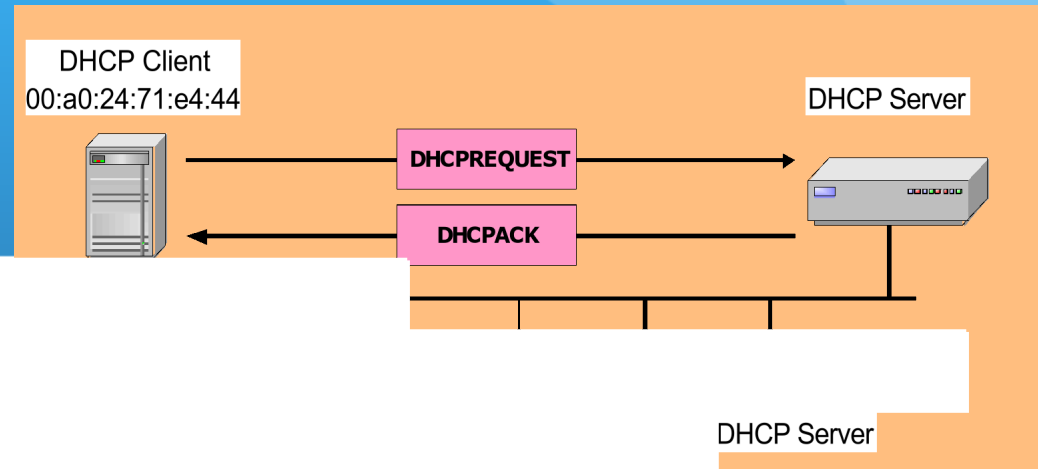
- The client broadcasts a DHCPDISCOVER message on its local physical subnet.
 - The DHCPDISCOVER message contains some options such as the client's MAC address and an available IP address.
- The servers record the address as offered to the client to prevent the same address being offered to other clients in the event of further DHCPDISCOVER messages being received before the first client has completed its configuration.

DHCP Operation - accepts offer from one server

- DHCP REQUEST
Accepts one offer

At this point the client can start address

- Renewing a Lease (sent when 50% of lease has expired)
- If DHCP server sends DHCPNACK, then address is released when timer expires



Contd.

- If the client receives one or more DHCPOFFER messages from one or more servers.

- The client chooses _____ configuration

IP a
selected offer.

- In the event that no offers are received, if the client has knowledge of a previous network address, the client may reuse that address if its lease is still valid, until the lease expires.

Contd.

- The **servers receive** the **DHCPREQUEST** broadcast from the client.
- Those **servers not selected** by the **DHCPREQUEST** message use the message as notification that the client has **declined** that server's offer.

DHCPA
the requesting client.

eters for

Contd.

- The combination of **client hardware** and **assigned network address** constitute a **unique identifier** for the client's lease and the client and server
- contain

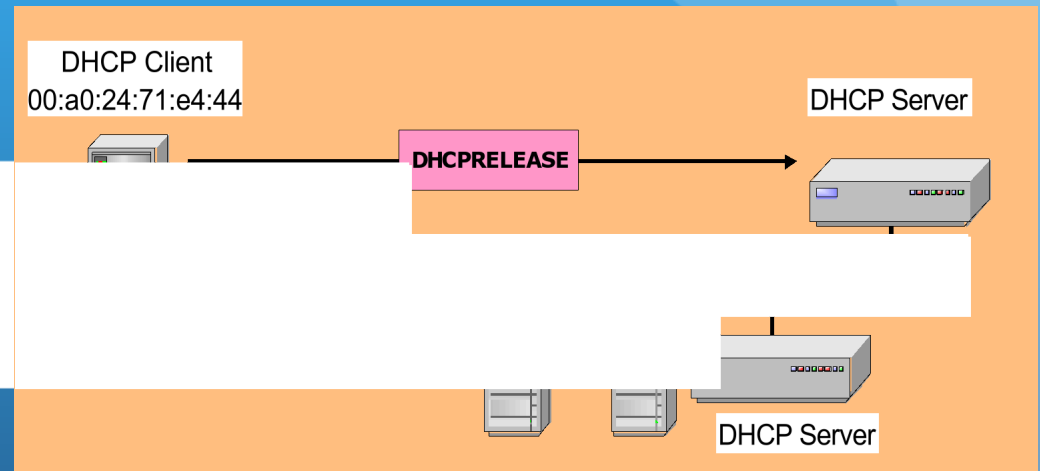
Contd.

- The **client receives** the **DHCPACK** message with configuration parameters.
 - The client performs the configuration parameters, for example, sets the IP address, subnet mask, and default gateway.
- If the client **detects a problem** with the parameters in the **DHCPACK** message (the address is already in use on the network, for example), the **client sends** a **DHCPDECLINE** message to the server and restarts the configuration process.

DHCP Operation - Release

- DCHP RELEASE

At t
client has relea
address



Contd.

- The **client** may choose to **relinquish** its lease on a network address by sending a **DHCPRELEASE** message to the server.

includ

ddress.

DHCP Pros

- It relieves the network administrator of a great deal of manual configuration work.
- The ability for a device to find its way to the network and to obtain an IP address from a central server.
- Because mobile clients are actually active, it is possible, by the use of reasonably short lease times and the fact that mobile clients do not need to be allocated more than one address, to reduce the total number of addresses in use in an organization.

DHCP Cons

- Uses UDP, an unreliable and insecure protocol.
- DNS cannot be used by untrusted hosts.

Domain Name Service - DNS



Outline

- What is DNS?
- What services d
- Mess
- Types of messages

What is DNS?

- DNS is a host name to IP address translation service
- DNS is
 - an intermediary between clients and servers
 - a name

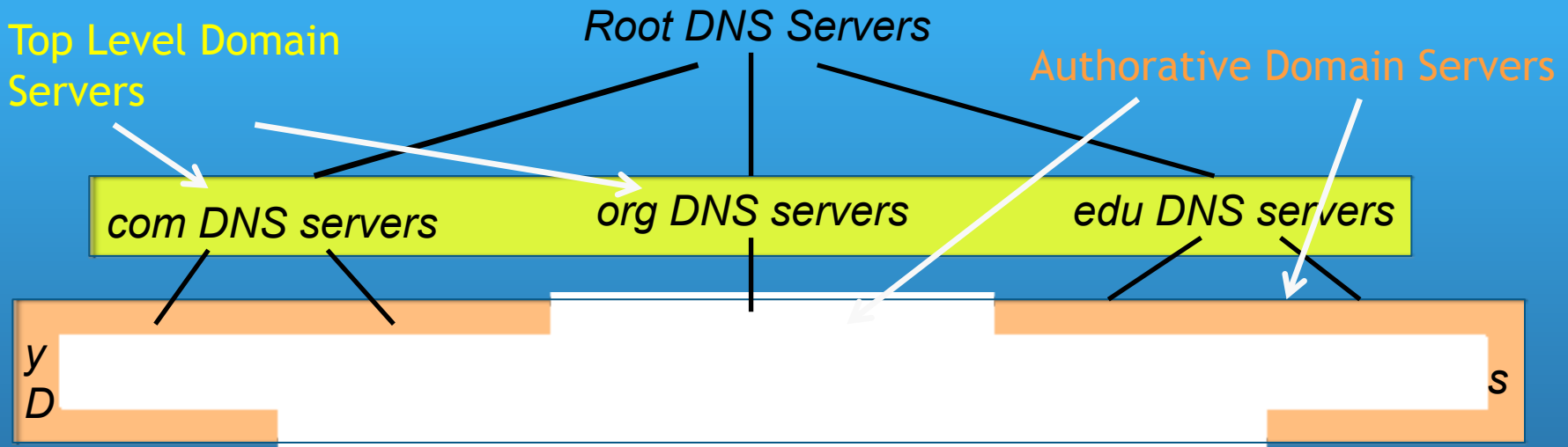
Why DNS?

- It is easier to remember a host name than it is to remember an IP address.
- A name has a user than a
- Appl , etc., all require the user to input a destination.
- The user generally enters a host name.
- The application takes the host name supplied by the user and forwards it to DNS for translation to an IP address.

How does it work?

- DNS works by exchanging messages between client and server machines.
- The application then sits and waits for the response to return.

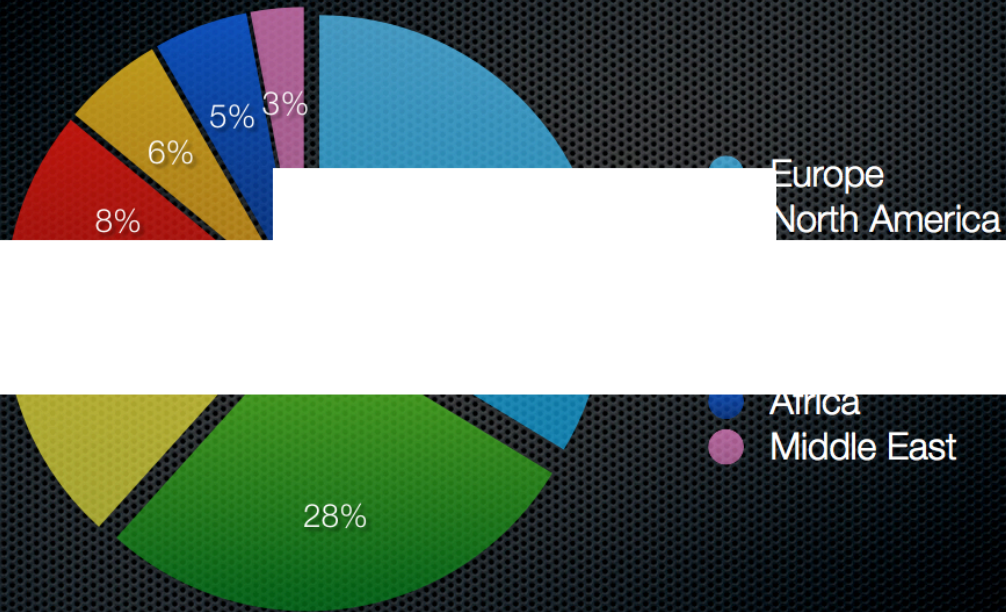
Distributed, Hierarchical Database



Client wants IP for www.amazon.com; 1st approx:

- client queries a root server to find “com” DNS server
- client queries “com” DNS server to get “amazon.com” DNS server
- client queries “amazon.com” DNS server to get IP address for “www.amazon.com”

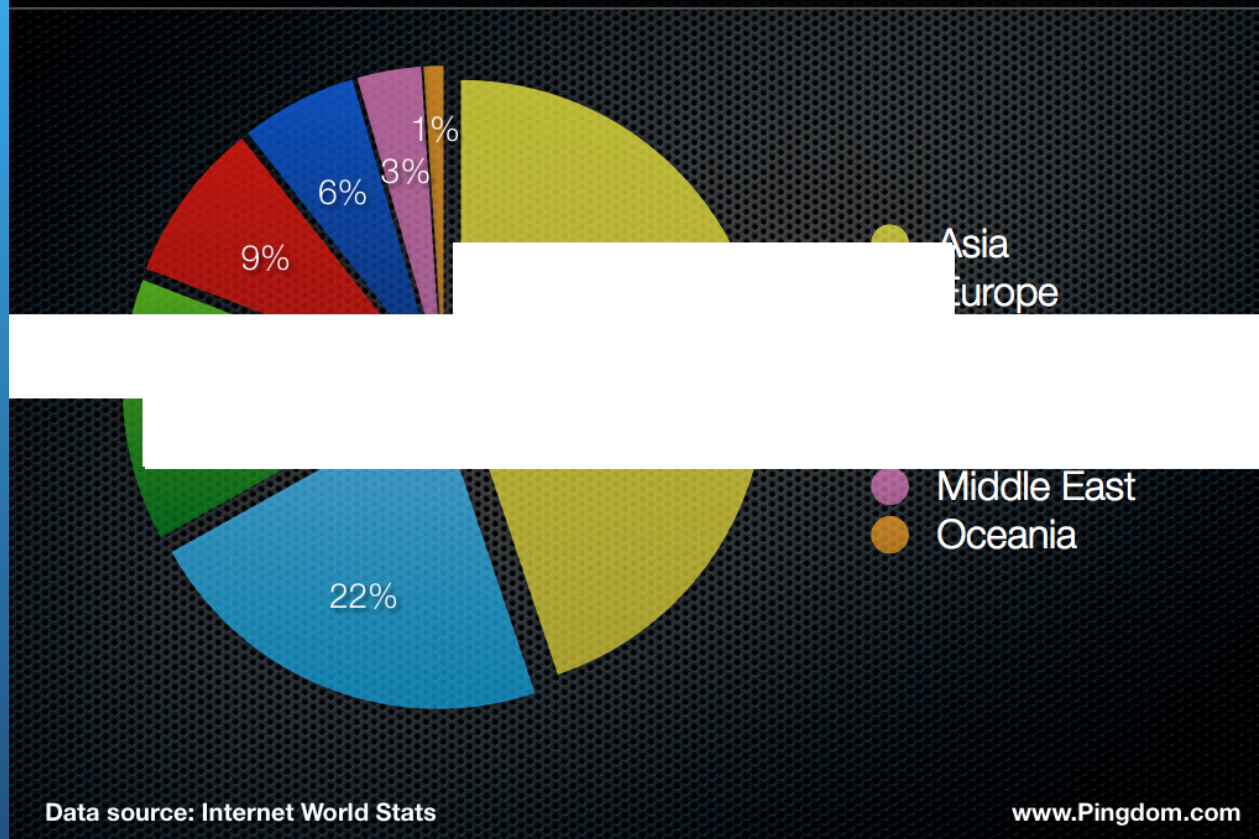
Geographical distribution of DNS root server sites



Data source: Root-servers.org

www.Pingdom.com

Geographical distribution of Internet users



DNS: Root name servers

- contacted by local name server that cannot resolve name
- root name server:
 - contacts authoritative name server if name mapping not known
 - gets mapping
 - returns mapping

VeriSign
US DoD
US DoD Vienna, VA
ARL Aberdeen, MD
VeriSign

NASA Mt View, CA
Internet Systems Consortium. Palo Alto, CA

USC-ISI Marina del Rey, CA
ICANN Los Angeles, CA

RIPE London

Autonomica, Stockholm

WIDE Tokyo

13 root name
server operators
worldwide

TLD and Authoritative Servers

- **Top-level domain (TLD) servers:**
 - responsible for com, org, net, edu, etc, and all top-level countries, ca, jp.
 - Educause for edu TLD
- **Authoritative DNS servers:**
 - organization's DNS servers, providing authoritative hostname to IP mappings for organization's servers (e.g., Web, mail).
 - can be maintained by organization or service provider

Local Name Server

- does not strictly belong to hierarchy
- each ISP (reside university) has
- when host makes DNS query, query is sent to its local DNS server
 - acts as proxy, forwards query into hierarchy

DNS Queries

- Recursive:

- The client machine sends a request to the local name server, which, in turn, sends a request to the authoritative name server in its database.

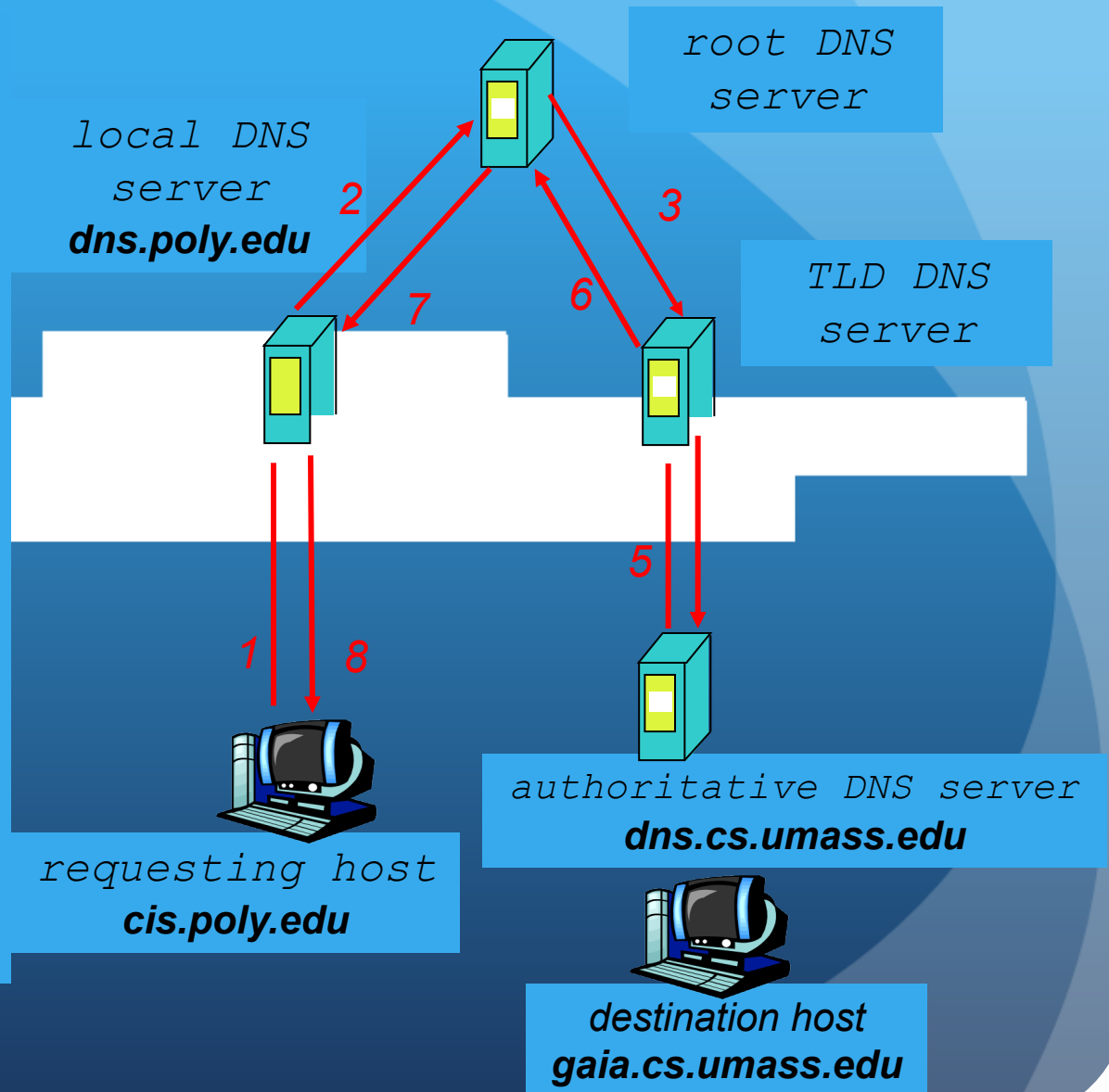
or
The local name server can contain some hostname to IP address mappings. The intermediate or Top Level name server always knows who the authoritative name server is.

DNS name resolution example

- Host at cis.poly.edu wants IP address for: gaia.cs.umass.edu

recursive query

- ❑ puts burden of name resolution on contacted root name server
- ❑ heavy load



DNS Queries (cont'd)

- Iterative:
 - The local server queries the root server. If address not in its database, address of an

dir
aut e
overloading of the root servers that handle millions of requests.

DNS name resolution example

- Host at cis.poly.edu wants IP address for gaia.cs.umass.edu

❑ contact
replies with name of
server to contact
-> "I don't know this
name, but ask this
server"

local DNS server

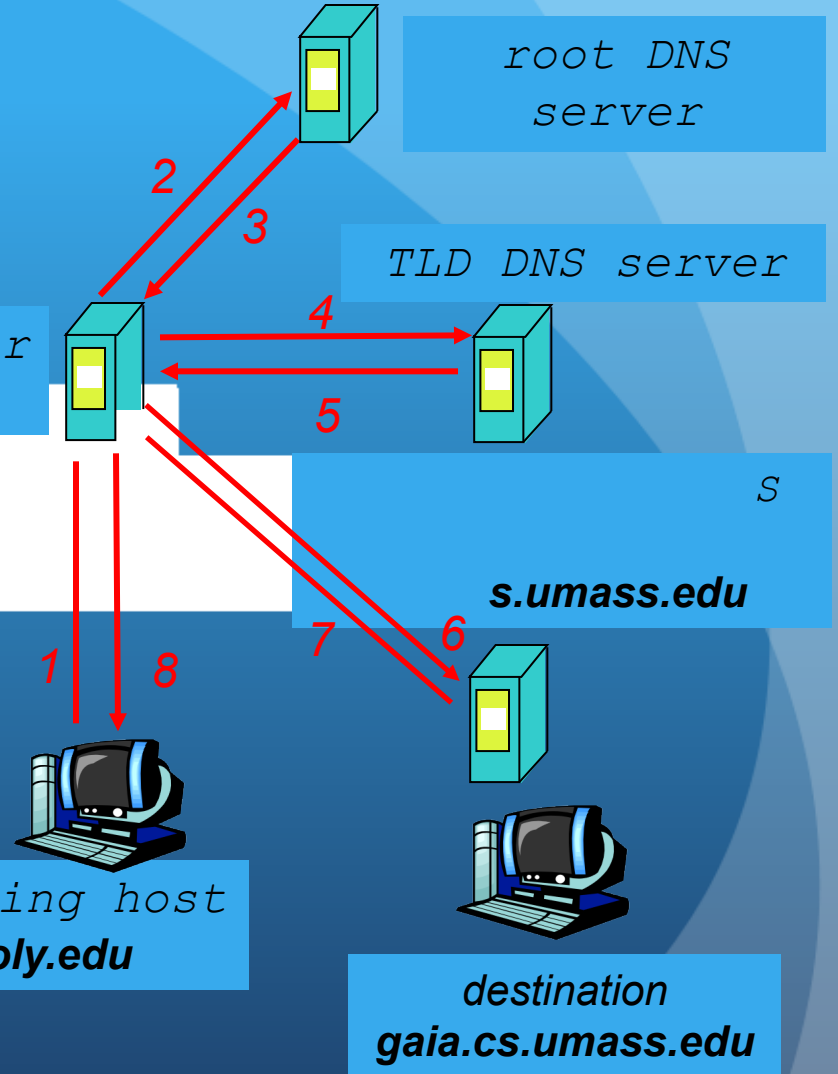
root DNS
server

TLD DNS server

s.umass.edu

requesting host
cis.poly.edu

destination
gaia.cs.umass.edu



DNS: caching and updating records

- once (any) name server learns a mapping, it *caches* mapping
 - cache entries time
 - TTL time

Operation of DNS

- The DNS data is stored in the database in the form of resource records (RR). The RRs are directly inserted in the

RRs

- TTL: time to live, used to indicate when an RR can be removed from the DNS cache.
- Type =

addre
 - CNAME – then NAME is an alias for a host and Value is the canonical name for the host
 - MX – then NAME is an alias for an email host and Value is the canonical name for the email server

DNS records

DNS: distributed db storing resource records (RR)

RR format: (name, value, type, ttl)

- **Type=A**

- **Type=NS**

- **name** is domain (eg., foo.com)
- **value** is hostname of authoritative name server for this domain

- **Type=CNAME**

is alias name for

really

servereast.backup2.ibm.com

- **value** is canonical name

- **Type=MX**

- **value** is canonical name of mailserver associated with name

Summary

- DNS provides a mechanism for maintaining the user friendliness of the Internet by hiding some of the operational data

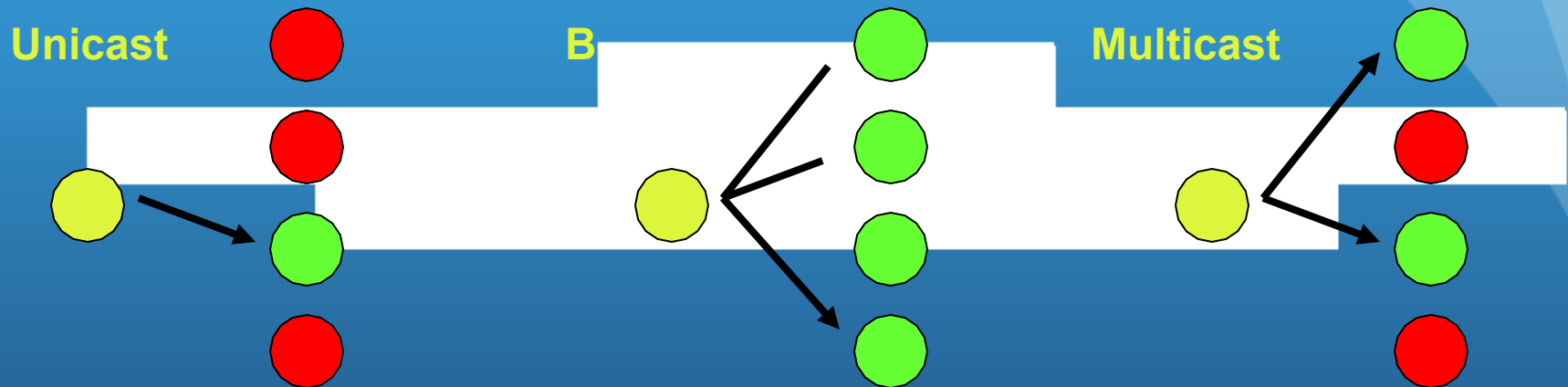
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IP Multicasting



Multicasting

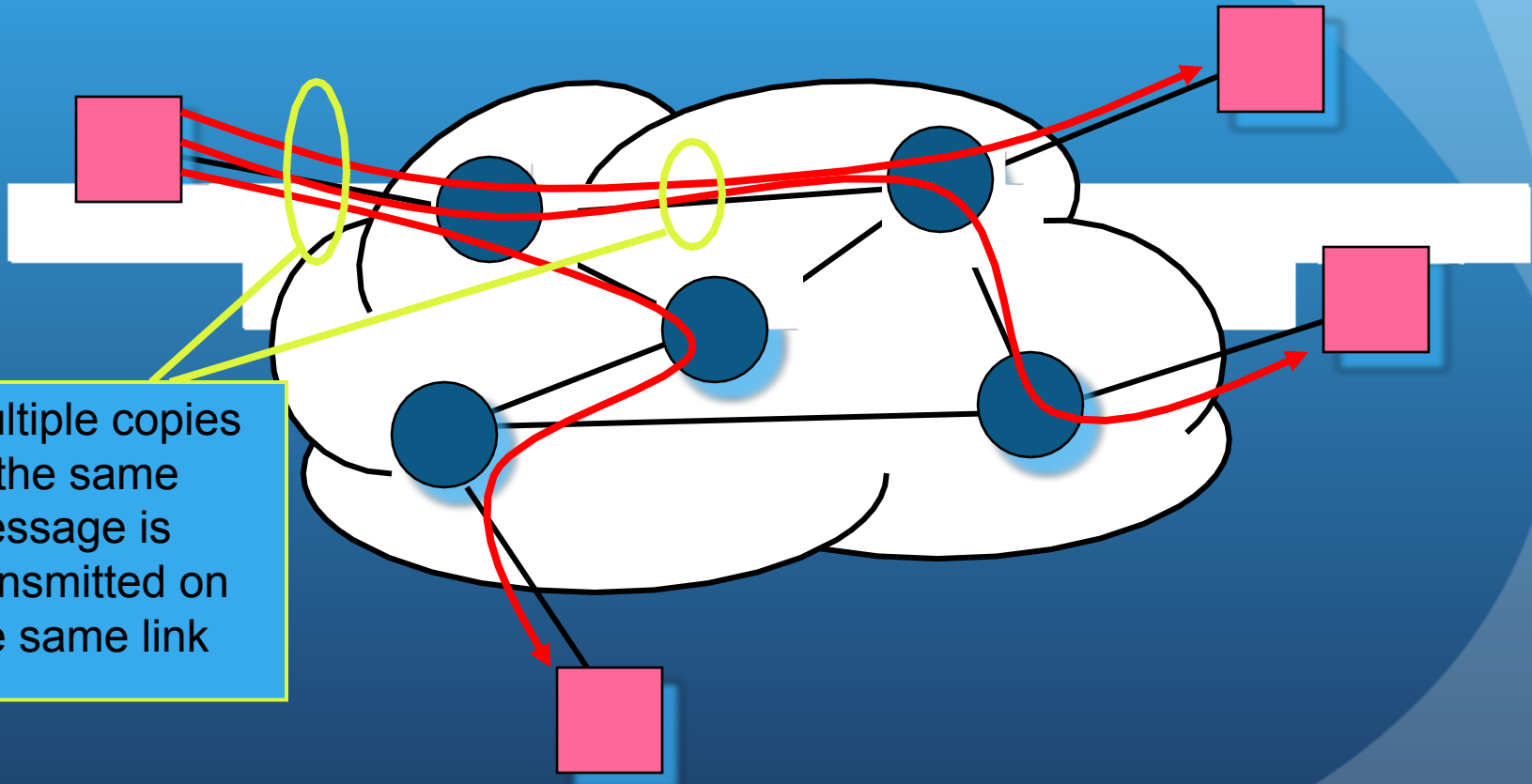
- Multicast communications refers to one-to-many or many-to-many communications.



IP Multicasting refers to the implementation of multicast communication in the Internet

Multicasting over a Packet Network

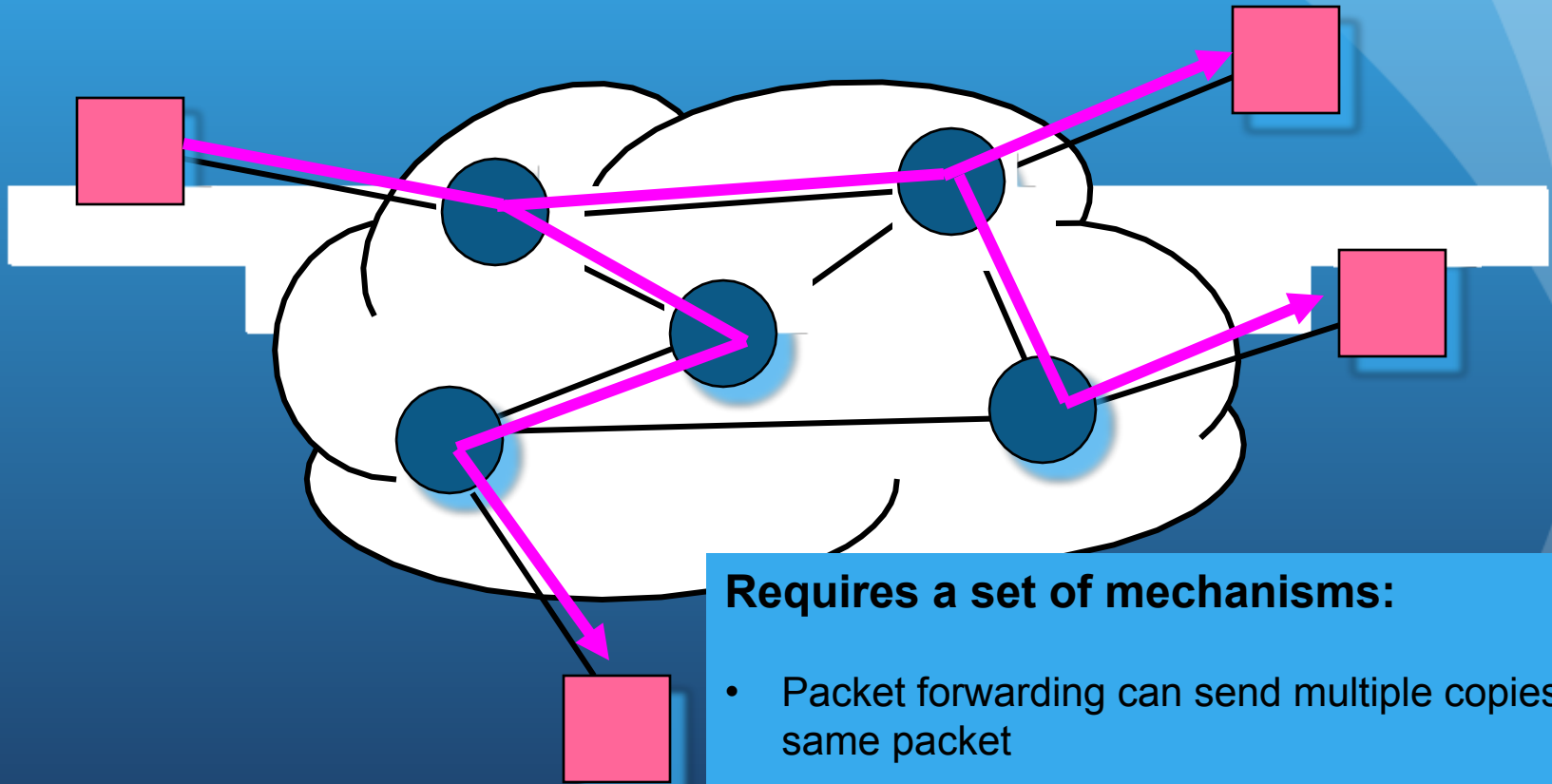
- Without support for multicast at the network layer:



Multiple copies of the same message is transmitted on the same link

Multicasting over a Packet Network

- With support for multicast at the network layer:



Requires a set of mechanisms:

- Packet forwarding can send multiple copies of same packet
- Multicast routing algorithm which builds a spanning tree (dynamically)

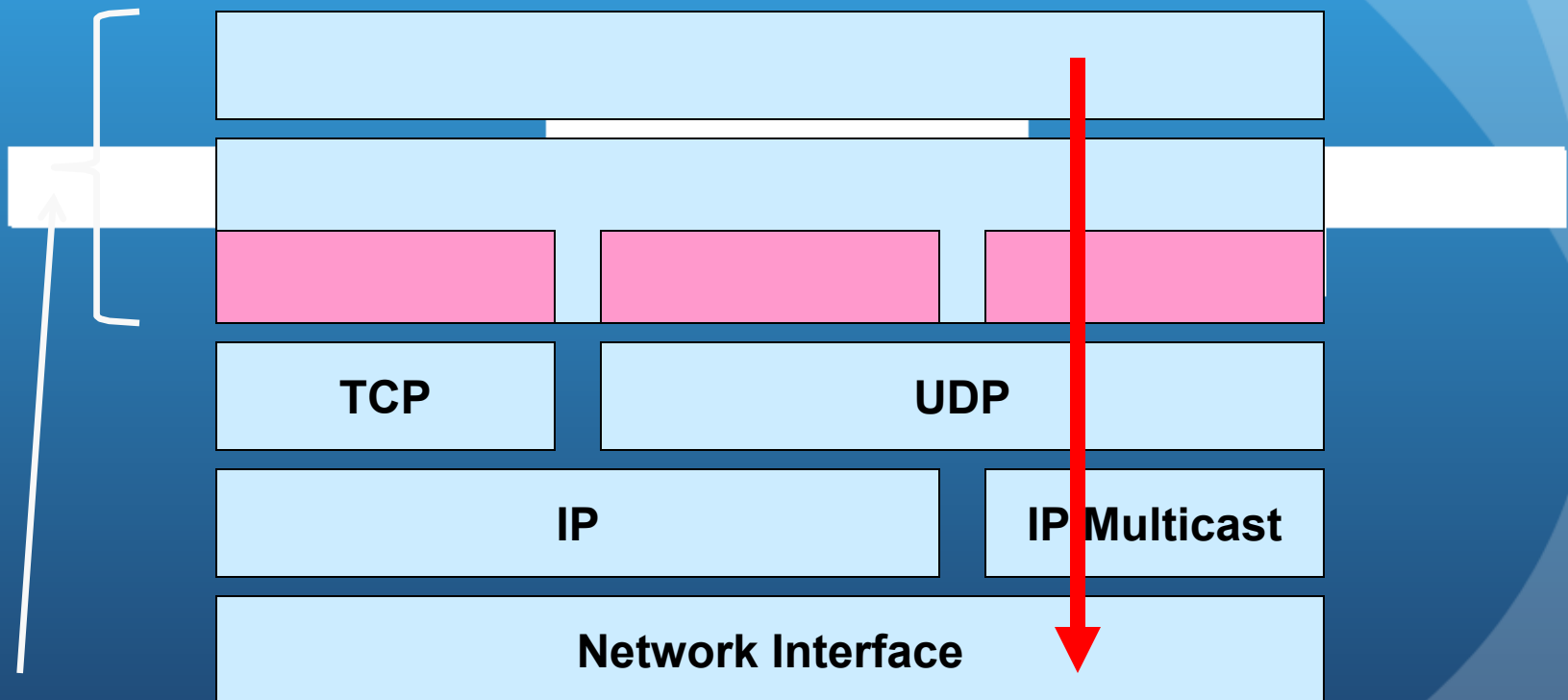
Semantics of IP Multicast

IP multicast works as follows:

- Multicast groups are identified by IP addresses in the range 224.0.0.0 - 239. (D address)
- Every IP datagram sent to a multicast group is transmitted to all members of the group
- The IP Multicast service is **unreliable**

The IP Protocol Stack

- IP Multicasting only supports UDP as higher layer
- There is no multicast TCP !



Application Layer

Multicast Addressing

- All multicast addresses start with (old class D addresses):

Class D

1	1	1	0
---	---	---	---

multicast group id

28 bits

- Multicast addresses are dynamically assigned.
- An IP datagram sent to a multicast address is forwarded to everyone who has joined the multicast group
- If an application is terminated, the multicast address is (implicitly) released.

Types of Multicast addresses

- The range of addresses between 224.0.0.0 and 224.0.0.255, inclusive, is reserved for the use of routing protocols and other low-level topology discovery or maintenance protocols

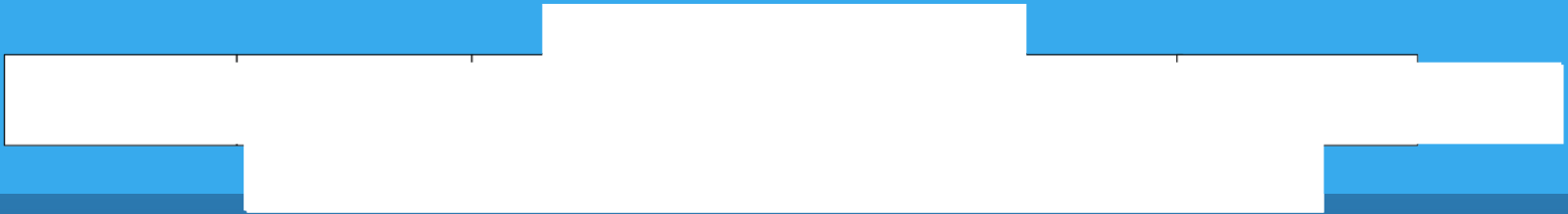
- Multicast address

- Examples

224.0.0.1	All systems on this subnet
224.0.0.2	All routers on this subnet
224.0.1.1	NTP (Network Time Protocol)
224.0.0.9	RIP-2 (a routing protocol)

Multicast Address Translation

- In Ethernet MAC addresses, a **multicast address** is identified by setting the lowest bit of the “most left byte”

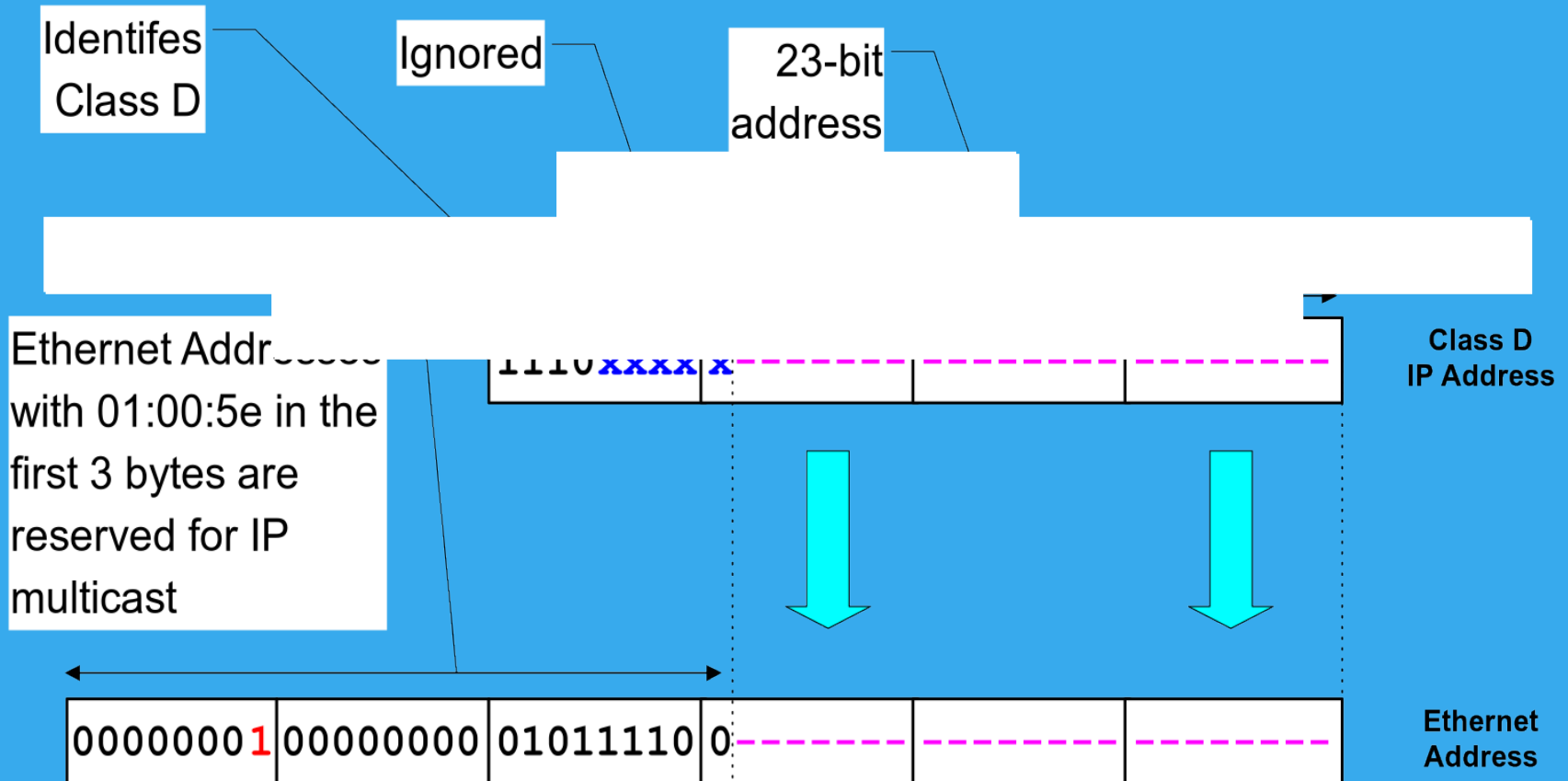


Not all Ethernet cards can filter multicast addresses in hardware

Then: **Filtering is done in software by device driver.**

Ethernet uses multicasting for various protocols such as spanning tree protocol or VLAN set up. IP Multicast is distinguished a special code in the 3rd octet.

IP Multicast Address Mapping



IGMP

- The Internet Group Management Protocol (IGMP) is a simple management protocol for the support of IP multicast.
- IGMP membership in a multicast group.
- Support for:
 - Joining a multicast group
 - Query membership
 - Send membership reports

Network Time Protocol - NTP



To Synchronize or not to synchronize

- Criticality of the situation - how does timing affect the outcome of an action or sequence of actions
- The ordering of events is done using a common clock
- Network Time Protocol allows for timing exchange to synchronize clocks.

NTP

- NTP is a protocol
- NTP is a set of ti

(strat

- Stratum “0” being the top and they are atomic clocks
- Stratum “1” are time servers connected to stratum “0”
- And Stratum “2” are connected to stratum “1” etc.
- Clients get information from their local time server at stratum “N”

NTP Daemon - ntpd

- On most systems, there is an ntpd daemon that synchronizes the local clock to a time server in the area. Often a per time server they
- NTP operates by getting the time from the local time server and estimating a clock offset to adjust its own clock.

NTP calculation

- Client A sends a packet to time server at time t_0 .
- Client A receives the server at time t_3
- Serve
- Network latency estimate = $((t_3 - t_0) - (t_2 - t_1))/2$
- Clock offset estimate = $(t_1 - t_0) - \text{Network latency estimate}$
- Packet from Server to Client contains t_0, t_1, t_2

NTP Packet Exchange

- UDP port 123
- Packets sent:
 - Client request with
 - R
 - Time of response to client - t_2
- Client can have several servers
- Client chooses one to sync with
- Uses feedback loop to keep running estimate of RTT and offset.